

GME: Improving Universal Multimodal Retrieval by Multimodal LLMs

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<https://hf.co/Alibaba-NLP/gme-Qwen2-VL-2B-Instruct>

Abstract

Universal Multimodal Retrieval (UMR) aims to enable search across various modalities using a unified model, where queries and candidates can consist of pure text, images, or a combination of both. Previous work has attempted to adopt multimodal large language models (MLLMs) to realize UMR using only text data. However, our preliminary experiments demonstrate that more diverse multimodal training data can further unlock the potential of MLLMs. Despite its effectiveness, the existing multimodal training data is highly imbalanced in terms of modality, which motivates us to develop a training data synthesis pipeline and construct a large-scale, high-quality fused-modal training dataset. Based on the synthetic training data, we develop the General Multimodal Embedder (GME), an MLLM-based dense retriever designed for UMR. Furthermore, we construct a comprehensive UMR Benchmark (UMRB) to evaluate the effectiveness of our approach. Experimental results show that our method achieves state-of-the-art performance among existing UMR methods. Last, we provide in-depth analyses of model scaling, training strategies, and perform ablation studies on both the model and synthetic data.

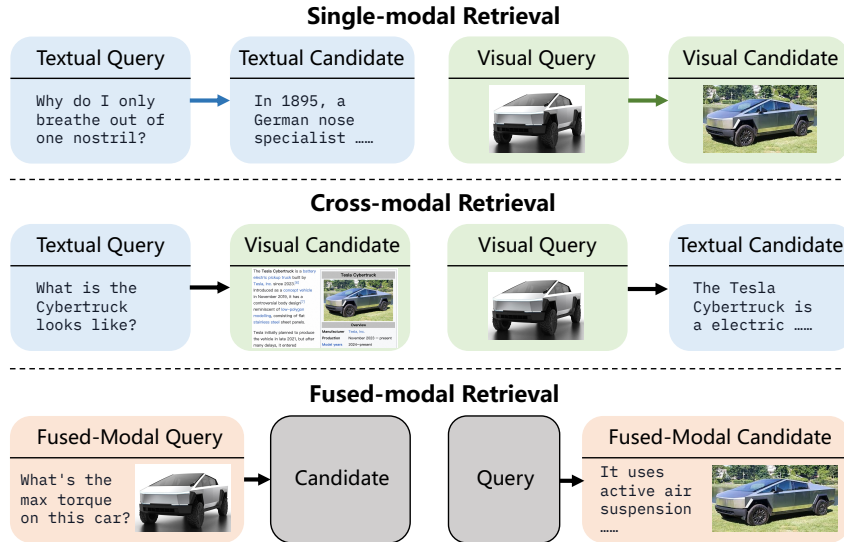


Figure 1: Illustration of different retrieval settings in our universal multimodal retrieval task. Blocks with black borders represent data in arbitrary modalities, *i.e.*, text-only, image-only or fused.

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Methods	Modeling		Retrieval Setting		
	Approach	Training	S&C	Fused	VD
UniVL-DR (Liu et al., 2023c)	CLIP Feature Fusion	Cross-modal	✓	✗	✗
UniIR (Wei et al., 2024)	CLIP Score Fusion	Multimodal	✓	✓	✗
	BLIP Feature Fusion				
MARVEL (Zhou et al., 2024b)	Text Encoder+Plugin	Cross-modal	✓	✗	✗
VISTA (Zhou et al., 2024a)	Text Encoder+Plugin	Multimodal	✓	✓	✗
E5-V (Jiang et al., 2024a)	MLLM	Text-only	✓	✓	✗
GME (Ours)	MLLM	Multimodal	✓	✓	✓

Table 1: Comparison of UMR studies. S&C, Fused, and VD denote the retrieval setting of single-modal & cross-modal, fused-modal, and retrieving visual documents (*e.g.*, PDF screenshots), respectively. The setting explanation is in Figure 1.

1 Introduction

The growth of multimedia applications necessitates retrieval models that extend beyond traditional text-to-text and text-to-image search (Zhou et al., 2024b). In Universal Multimodal Retrieval (UMR) tasks, both queries and candidates can exist in any modality (Liu et al., 2023c). Compared to addressing this challenge with separate unimodal and cross-modal retrievers in a divide-and-conquer pipeline (Chang et al., 2022), a unified retriever is a more viable option in terms of usability and scalability. Using the dense retrieval paradigm (also known as embedding-based retrieval) (Karpukhin et al., 2020), a unified model can be trained to project inputs from various modalities into a shared embedding space (Zhou et al., 2024b;a; Jiang et al., 2024a). In this space, similarity scores are computed between the embeddings of queries and the retrieval collection, facilitating the efficient ranking of the top- k candidates. To achieve this, some previous studies have primarily focused on two approaches: (1) designing feature fusion mechanisms for cross-modal retrievers based on the CLIP architecture (Liu et al., 2023c; Wei et al., 2024), and (2) incorporating visual plugin modules into optimized text embedding models to achieve unified multimodal representations (Zhou et al., 2024b;a).

Recently, researchers have turned to exploring Multimodal Large Language Models (MLLMs) (Liu et al., 2023a; Wang et al., 2024b) in UMR. For example, it is shown that training MLLMs with text data alone can generate universal multimodal embeddings with respectable retrieval performance (Jiang et al., 2024a). However, modality-limited training may fail to fully demonstrate the potential of MLLMs in UMR. We believe that incorporating multimodal data composition (as shown in Figure 1) could further enhance the model performance and generalization. Moreover, visual documents (*i.e.*, document screenshots) are increasingly important in UMR tasks, as they not only simplify the pipelines of diverse Retrieval-Augmented Generation (RAG) applications, but also mitigate information loss during modality conversion (Ma et al., 2024; Faysse et al., 2024b). However, current UMR models primarily target natural images, neglecting support for this scenario (Table 1).

To address the aforementioned challenges, we propose the General Multimodal Embedder (GME), an instruction-based embedding framework utilizing MLLMs as the backbone. GME enables retrieval across various modalities in the unified paradigm, including text, images, visual documents, and fused-modal¹ (*i.e.*, image-text composed) contents. Our framework is underpinned by two key techniques: (1) A strategically optimized training data composition for UMR. We categorize UMR tasks into three distinct types: single-modal, cross-modal, and fused-modal (Figure 1). Through extensive experimentation, we analyze how different compositions affect performance (Figure 3) and demonstrate that a balanced mixture of all types yields optimal results. (2) An efficient data synthesis pipeline for large-scale fused-modal training. Recognizing the under-representation of fused-modal data and its potential impact on training effectiveness, we develop a streamlined data synthesis

¹We use fuse-modal instead of multimodal to denote the data that contains both text and image to disambiguate from the UMR task.

pipeline (§4.2). This approach has successfully generated a comprehensive dataset of 1.1M fused-modal pairs, significantly enhancing our training and model capabilities.

To evaluate the effectiveness of our framework, we compile a comprehensive UMR Benchmark, namely **UMRB**. This benchmark encompasses tasks from widely recognized retrieval benchmarks in text (Thakur et al., 2021), multimodal (Wei et al., 2024), and visual document retrieval (Faysse et al., 2024b), as well as our newly processed fused-modal retrieval data. We build our models on top of the strong Qwen2-VL series MLLMs (Wang et al., 2024b) and train them on our constructed dataset. Experimental results demonstrate that our model achieves state-of-the-art performance on the UMRB benchmark. Additionally, we perform in-depth analyses on model scaling, training strategies, and ablation of our synthetic data. Our key contributions are:

- We explore strategies to adapt MLLMs into UMR models, and present GME, a powerful embedding model capable of retrieving candidates across different modalities. GME is the first UMR model to deliver visual document retrieval performance on par with specialized models.
- We propose a novel data synthesis pipeline for constructing large-scale, fused-modal training data to encounter the scarcity of such training data. This pipeline is more efficient than previous approaches and can be easily extended to other domains.
- We compile the UMR Benchmark, UMRB, to evaluate a broader range of retrieval tasks compared to existing benchmarks. UMRB categorizes tasks into three types: single-modal, cross-modal, and fused-modal, and offers a comprehensive performance evaluation across them.

2 Related Work

Multimodal Large Language Models The emergence of Large Language Models (LLMs) has driven significant progress in natural language processing (Brown et al., 2020; OpenAI, 2023b), leading to the development of Multimodal LLMs that extend these capabilities to handle multimodal information. Prominent MLLMs such as GPT-4V (OpenAI, 2023a), LLaVa (Liu et al., 2023a; 2024), Qwen-VL (Wang et al., 2024b), InternVL (Gao et al., 2024; Chen et al., 2024) and MiniCPM-V (Yao et al., 2024) have shown promising advancements in multimodal information understanding and reasoning. Typically, an MLLM consists of an LLM, a vision encoder, and a projector that bridges the two components by transforming raw multimodal inputs into vectors compatible with the LLM (Yin et al., 2023).

Multimodal Retrieval Early multimodal retrieval tasks focused on single-modal (Zhao et al., 2024) or cross-modal retrieval (Wang et al., 2016). Recently, the expansion of multimedia applications and multimodal retrieval-augmented generation (RAG) by MLLMs has created a need for unified multimodal retrieval models for complex scenarios. Existing approaches largely utilize pre-trained models such as CLIP (Radford et al., 2021) or BLIP (Li et al., 2022) for multimodal embedding. For instance, UniVL-DR (Liu et al., 2023c) and UniIR (Wei et al., 2024) initially encode images and texts separately using CLIP or BLIP encoders, followed by fusion strategies like score fusion to integrate features from both modalities. Additionally, VISTA (Zhou et al., 2024a) and MARVEL (Zhou et al., 2024b) employ pre-trained text embedding models enhanced with visual plugins to encode composite image-text candidates. However, these methods are typically designed for specific tasks like multimodal document retrieval and lack the flexibility to handle diverse multimodal retrieval tasks.

Concurrent with our work, E5-V (Jiang et al., 2024a) and VLM2VEC (Jiang et al., 2024b) propose fine-tuning MLLMs on single-text (NLI (Gao et al., 2021)) or vision-centric relevance data, demonstrating their transferability to multimodal retrieval. In this paper, we are the first to explore the fine-tuning of an MLLM-based universal multimodal retriever that can address both visual retrieval tasks and maintain strong text-to-text retrieval capabilities. Moreover, we are the first to extend a unified retrieval model to handle not only **natural image retrieval** but also **text-rich image retrieval** (Faysse et al., 2024b).

Class	Task	Datasets
Single-Modal (17)	T→T (16)	ArguAna (Wachsmuth et al., 2018) Climate-FEVER (Diggelmann et al., 2020) CQADupStack (Hoogeveen et al., 2015) DBpedia (Hasibi et al., 2017) FEVER (Thorne et al., 2018) FiQA2018 (Maia et al., 2018) HotpotQA (Yang et al., 2018) MSMARCO (Nguyen et al., 2016) NFCorpus (Boteva et al., 2016) NQ (Kwiatkowski et al., 2019) Quora ² SCIDOC5 (Cohan et al., 2020) SciFact (Wadden et al., 2020) Touche2020 (Bondarenko et al., 2020) TRECCOVID (Voorhees et al., 2020) WebQA (Chang et al., 2022)
	I→I (1)	Nights (Fu et al., 2023)
Cross-Modal (18)	T→I (4)	VisualNews (Liu et al., 2021a) Fashion200k (Han et al., 2017) MSCOCO (Lin et al., 2014) Flickr30k (Plummer et al., 2015)
	T→VD (10)	TAT-DQA (Zhu et al., 2022) ArxivQA (Li et al., 2024) DocVQA (Mathew et al., 2021) InfoVQA (Mathew et al., 2022) Shift Project [†] Artificial Intelligence [†] Government Reports [†] Healthcare Industry [†] Energy [†] TabFQuad [†]
	I→T (4)	VisualNews (Liu et al., 2021a) Fashion200K (Han et al., 2017) MSCOCO (Lin et al., 2014) Flickr30k (Plummer et al., 2015)
Fused-Modal (12)	T→IT (2)	WebQA (Chang et al., 2022) EDIS (Liu et al., 2023b)
	IT→T (5)	OVEN (Hu et al., 2023) INFOSEEK (Chen et al., 2023)
	IT→I (2)	ReMuQ (Luo et al., 2023) OKVQA (Marino et al., 2019) LLaVA (Lin et al., 2024)
	IT→IT (3)	FashionIQ (Wu et al., 2021) CIRR (Liu et al., 2021b)
		OVEN (Hu et al., 2023) EVQA (Mensink et al., 2023) INFOSEEK (Chen et al., 2023)

Table 2: An overview of tasks and datasets in our UMRB. † means that they all originate from Faysse et al. (2024a).

Embedding Models with Pre-trained Language Models With the advancement of pre-trained Language Models, research in both pure text and Vision-Language Models has focused on building representation models based on these pre-trained language models. In the text retrieval domain, state-of-the-art text embedding models such as Contriver (Izacard et al., 2021), E5 (Wang et al., 2022), GTE (Li et al., 2023), and BGE (Xiao et al., 2024) are all built upon pre-trained language models and have demonstrated impressive generalizability and robust performance in text retrieval tasks. Recently, leveraging LLMs combined with supervised fine-tuning (SFT), researchers have developed unified text representation models that fully utilize the text understanding capabilities of LLMs, resulting in models with enhanced performance and generalization (Wang et al., 2024a; Li et al., 2023; Lee et al., 2024). These models typically process user text inputs through LLMs, using the hidden states from the final transformer layer—either through pooling or by selecting the last token—as the final representation. Inspired by the success of universal text embedding models based on text LLMs, researchers have begun to explore the construction of unified multimodal retrieval models using MLLMs (Jiang et al., 2024a;b). In this paper, we aim to demonstrate through systematic experiments that constructing a truly universal multimodal retrieval model using MLLMs is feasible.

3 Universal Multimodal Retrieval

Current UMR sub-tasks can be categorized into three types based on the modalities of the query and the candidate:

- **Single-Modal Retrieval:** Both the query and the candidate belong to the same modality, such as text-to-text (T→T) or image-to-image (I→I) retrieval scenarios.
- **Cross-Modal Retrieval:** The query and the candidate belong to different modalities, typically text-to-image (T→I) retrieval. Unlike most prior work that focuses on natural-style image retrieval, we also consider the retrieval of rich-text images (e.g., images converted from scholarly PDFs). We denote this scenario as text-to-visual document (T→VD) retrieval.
- **Fused-Modal Retrieval:** More complicated retrieval tasks involve mixed modalities in queries, candidates, or both. For example, in EVQA (Mensink et al., 2023), both queries and candidates combine text and images.

The visualization of these settings refers to Figure 1.

²More details about the dataset can be found at Quora Dataset Release: Question Pairs.

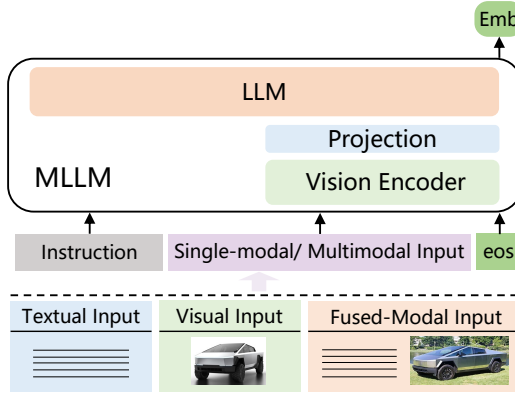


Figure 2: The GME model architecture. Emb denotes the embedding of the input content.

3.1 Universal Multimodal Retrieval Benchmark

Based on the aforementioned classification principles, we introduce a new benchmark to comprehensively assess the performance of UMR models. This benchmark comprises 47 evaluation datasets that cover a broad spectrum of multimodal retrieval tasks, and we name it the Universal Multimodal Retrieval Benchmark (UMRB). These evaluation datasets primarily originate from previously constructed datasets tailored for each sub-scenario or sub-task. Specifically, UMRB includes: (1) The BEIR (Thakur et al., 2021) benchmark for text-to-text retrieval scenarios; (2) The M-BEIR (Wei et al., 2024) dataset for vision-centric retrieval scenarios; (3) Additional fused-modal datasets that not cover by M-BEIR; and (4) text-to-visual document search datasets, such as ViDoRe (Faysse et al., 2024b), to extend the coverage of our benchmark and ensure a comprehensive evaluation of model universality. A detailed list of the UMRB datasets is presented in Table 2.

Given the extensive size of UMRB, to expedite our experimental validation and analysis, we have sampled a subset of datasets from each category, constituting a smaller dataset named UMRB-Partial. This subset retains 39% of the total datasets while maintaining evaluation richness. More detailed statistical information about UMRB-Partial can be found in our supplementary materials.

4 Method

In this section, we present the training framework for developing the General Multimodal Embedder (GME) model using a Multimodal Large Language Model (MLLM). We describe the contrastive learning approach used to train the embedding model. Building on this, we conduct detailed experiments to determine the optimal balance of training data types. Specifically, our experiments demonstrate that diverse data type mixtures significantly enhances the model’s ability to perform retrieval across various modalities. Lastly, recognizing the scarcity of high-quality fused-modal training data, we propose a novel method for automatically synthesizing large-scale, high-quality training data using MLLM.

4.1 GME: General Multimodal Embedder

Model Architecture We employ a MLLM as the foundation for GME. This model can accept images, text, or image-text pairs as input. Inspired by previous research on text embedding (Li et al., 2023; Wang et al., 2024a), we use the **final hidden state of the last token** as the representation (or embedding) for the input. Although pre-trained MLLMs possess strong multimodal understanding capabilities, their original training objectives are not optimized for representation learning. Therefore, task-specific fine-tuning (or alignment) is necessary to enhance the model’s representational capacity. Contrastive learning has been

shown to effectively train LLMs and MLLMs to produce retrieval embeddings (Li et al., 2023; Jiang et al., 2024a).

Contrastive Learning In our contrastive learning setup, each training instance comprises a query q , a relevant candidate c , and a set of irrelevant candidates $\{c_1^-, c_2^-, \dots, c_K^-\}$. Both q and c can be text, images, or image-text pairs, allowing the model to handle diverse data modalities. To tailor the model to various downstream retrieval tasks, we incorporate an instruction tuning method by including a tailored instructional text i with each retrieval task. For example, for the Visual Question Answering (VQA) task, the instruction could be: “Retrieve a passage that provides an answer to the given query about the image” guiding the model on how to process and interpret the query for specific objectives.

During training, we input q and instruction i into the model to obtain the query representation e_q . Similarly, each candidate c is input into the model to obtain its representation e_c . The training objective minimizes the cosine distance between e_q and e_c for relevant pairs while maximizing the distance between e_q and e_{c^-} for irrelevant pairs. Cosine similarity is employed to measure the directional alignment between embeddings, effectively capturing semantic similarities irrespective of their magnitudes.

The optimization process utilizes the InfoNCE loss function (van den Oord et al., 2018), defined as:

$$\mathcal{L} = -\log \frac{\exp(\cos(e_q, e_c^+)/\tau)}{\exp(\cos(e_q, e_c^+)/\tau) + \sum_{i=1}^K \exp(\cos(e_q, e_{c_i^-})/\tau)} \quad (1)$$

where τ is the temperature parameter that scales the cosine similarities to control the distribution’s concentration. This approach ensures that the model effectively learns to distinguish relevant from irrelevant information across different modalities, thereby enhancing its performance in multimodal retrieval tasks.

Hard Negatives The quality and diversity of negative samples are essential for improving contrastive learning models (Robinson et al., 2020). Inspired by ANCE (Xiong et al., 2021), we employ a two-stage training strategy: (1) Initial Training: We first train the model using randomly selected negative candidates, resulting in Model M_1 . (2) Hard Negative Mining and Continue Training: Using M_1 , we retrieve the top K candidates for each query and select non-relevant candidates from them as hard negatives. We then use these hard negatives to further train M_1 , refining it into the final model. This approach ensures the model learns from both easily distinguishable and more challenging examples, thereby enhancing overall performance.

Training Data Composition A critical factor in multimodal representation learning is the composition of training data. Although previous studies like (Jiang et al., 2024a) have demonstrated that MLLMs can develop multimodal representation capabilities after being fine-tuned on single-modal data, the effect of data diversity on model performance remains unclear. Therefore, we compare the performance of models trained with different data combinations across various retrieval scenarios within our classification principle. Specifically, we used four types of training data: single-modal (including T→T and I→I), cross-modal (including T→VD and T→I), fused-modal training data (including IT→IT), and a mixed dataset combining the first three types. These different training data types result in a total of six models.

For single-modal data, we utilized the T→T dataset from MS MARCO (Nguyen et al., 2016) and the I→I dataset from ImageNet (Deng et al., 2009), treating **images within the same category as positive matches and those from different categories as negatives**. For cross-modal data, we employed T→I pairs from the LAION (Schuhmann et al., 2022) dataset and T→VD pairs from the Docmatix (Laurençon et al., 2024) dataset. For fused-modal data, we use the EVQA (Mensink et al., 2023) dataset (IT→IT). For each subcategory, we randomly sampled 100,000 training instances to train the models independently. For the mixed dataset, we uniformly sampled 20,000 instances from each of the five datasets to train the final model,

Single-Modal	50.3	39.1	44.9	45.2	45.1	51.1
Cross-Modal	67.7	56.8	75.5	73.8	60.2	78.4
Fused-Modal	48.2	41.5	42.7	45.7	49.3	51.9
All	55.4	45.8	54.4	54.9	51.6	60.4
	T → T	I → I	T → VD	T → I	IT → IT	Mix

Figure 3: Impact of training data on multimodal retrieval tasks.

ensuring fair and reliable comparative experimental results. The performance of these six models on the UMRB-Partial test dataset is presented in Figure 3.

The results indicate that: (1) Models trained on single data types excel in corresponding retrieval tasks. For instance, models trained on T→T data performed best in text retrieval tasks.³ (2) A balanced mix of different data types enhanced performance across various settings. This suggests that increasing the diversity of training modalities effectively improves the model’s overall retrieval capabilities.

The above analysis highlights the importance of adequately representing each data type in training datasets to develop models that meet the requirements of universal multi-modal retrieval. During data collection, we observed that single-modal and cross-modal data are abundant, with over ten million training instances available. In contrast, fused-modal data remains limited. Common fused-modal training datasets such as EVQA(Mensink et al., 2023), INFOSEEK(Chen et al., 2023), and CIRRR (Liu et al., 2021b) collectively contain fewer than one million instances. Additionally, these existing fused-modal datasets cover only a limited range of domains. Thus, efficiently supplementing high-quality fused-modal training data is essential. To address this challenge, we propose leveraging the generative capabilities of LLMs and MLLMs to synthesize additional training data.

4.2 Fused-Modal Data Synthesis

To efficiently synthesize high-quality data while minimizing manual intervention, we adopt a strategy similar to Doc2Query (Gospodinov et al., 2023). However, our approach differs in that we aim to generate fuse-modal candidate-to-query relevance data instead of single-modality, text-based relevance pairs. This requires obtaining high-quality candidates that include both image and text content. We primarily extracted such data from Wikipedia paragraphs⁴. Additionally, to enhance the domain diversity of the candidate data, we employed a domain classification model⁵ to perform fine-grained classification of Wikipedia data into categories such as animals and plants. We then uniformly sampled from these categories and retained data with classification confidence scores above 0.5. Ultimately, we obtained 313,284 candidate entries, each containing both text and image content.

Based on these prepared candidate data, as shown in Figure 4, the overall data synthesis pipeline can be divided into the following steps: **Doc2Query Generation:** The passage content from each candidate is input into an LLM⁶ using a prompt to generate a natural query based on the passage. To ensure the quality of the generated queries, we built a

³Detail results are shown in the supplementary materials.

⁴WikiWeb2M

⁵We use bart-large-mnli as the classifier.

⁶In the entire synthesis process, we utilize Qwen/Qwen2.5-72B-Instruct as our LLM.

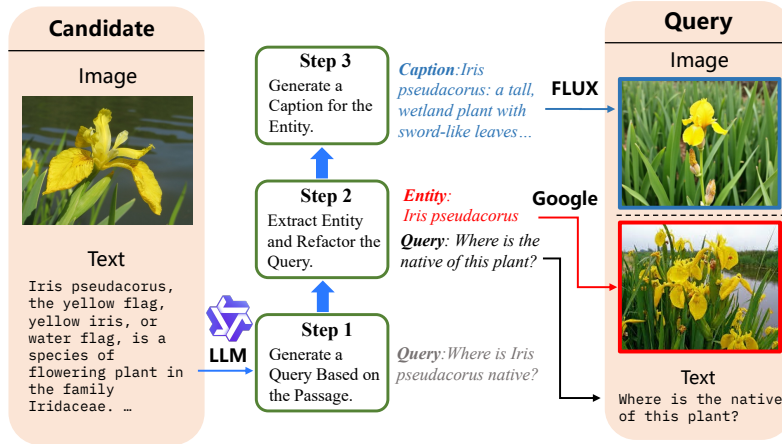


Figure 4: Pipeline for synthesizing fused-modal training data.

vector index of all passage contents using a text vector retrieval model⁷. Each generated query is then used to retrieve the corresponding passage from this collection. If the passage associated with the query is not within the top 20 retrieved items, the query is considered low quality due to low relevance and is discarded. In this step, we discarded 1.2% of the total generated queries. This process allows us to construct T→IT training data. **Entity Extraction and Query Rewrite:** We aim for the synthesized queries to include both texts and images (i.e., IT→IT type). To achieve this, we leverage entity extraction followed by image retrieval for the extracted entities and caption generation to supplement the image data on the query side. Specifically, for each generated query q from the first step, we use an Entity Extraction approach with an LLM to extract entities and rewrite the original query into q' . For example, the query “Where is *Iris pseudacorus* native?” is transformed by the model to the rewritten query “Where is the native habitat of this plant?” with the entity “*Iris pseudacorus*” extracted. We then seek images that match this entity and combine them with the rewritten query q' to form the final fuse-modal query. **Image Retrieval and Generation:** We explore two methods for obtaining images. The first method uses the Google Image Search API⁸ to retrieve images matching the entity terms, retaining the top five results. The second method involves generating images using a text-to-image model⁹. Specifically, we first use the LLM to generate a caption suitable for image generation based on the entity and the passage of the generated query, then input this caption into the text-to-image generation model to create the corresponding image. This approach allows us to quickly and efficiently obtain high-quality, diverse images. The synthesized results can also be assembled into IT→IT retrieval type data. **Data Filtering:** To ensure the quality of the synthesized data, we performed filtering on the final dataset. We observed that images generated by the FLUX model had consistent quality, whereas images retrieved via the Google Image Search API often included noisy data. Therefore, for images obtained through the Google Image Search API, we used the CLIP model¹⁰ to assess image-caption relevance. Images with a relevance score below 0.2 were filtered out.

Through the aforementioned synthesis pipeline, we produced 1,135,000 high-quality fuse-modal training data entries (including T→IT and IT→IT types). After filtering, we retained 1,102,000 entries, resulting in a data loss rate of 2.9%. The entire process consumed 600 A100 GPU hours. Detailed descriptions of all prompts used in the data synthesis pipeline and examples of the synthesized data will be provided in the Appendix §D.

⁷We use Alibaba-NLP/gte-Qwen2-1.5B-instruct as the retriever.

⁸<https://serpapi.com/google-images-api>

⁹<https://huggingface.co/black-forest-labs/FLUX.1-dev>

¹⁰[openai/clip-vit-large-patch14](https://openai.com/clip-vit-large-patch14)

UMRB	Size	Single-Modal		Cross-Modal				Fused-Modal				Avg.
Task (#Datasets)		T→T (16)	I→I (1)	T→I (4)	T→VD (10)	I→T (4)	T→IT (2)	IT→T (5)	IT→I (2)	IT→IT (3)	(47)	
VISTA (2024a)	0.2B	55.15	31.98	32.88	10.12	31.23	45.81	53.32	8.97	26.26	37.32	37.32
CLIP-SF (2024)	0.4B	39.75	31.42	59.05	24.09	62.95	66.41	53.32	34.90	55.65	43.66	43.66
One-Peace (2023)	4B	43.54	31.27	61.38	42.9	65.59	42.72	28.29	6.73	23.41	42.01	42.01
DSE (2024)	4.2B	48.94	27.92	40.75	78.21	52.54	49.62	35.44	8.36	40.18	50.04	50.04
E5-V (2024a)	8.4B	52.41	27.36	46.56	41.22	47.95	54.13	32.90	23.17	7.23	42.52	42.52
GME-Qwen2VL-2B	2.2B	55.93	29.86	57.36	87.84	61.93	76.47	64.58	37.02	66.47	64.45	64.45
GME-Qwen2VL-7B	8.2B	58.19	31.89	61.35	89.92	65.83	80.94	66.18	42.56	73.62	67.44	67.44

Table 3: Results of different models on our benchmark. Following previous works (Thakur et al., 2021; Wei et al., 2024; Faysse et al., 2024b), we present NDCG@10 scores for T→T tasks, excluding the WebQA dataset. For T→VD tasks, we provide NDCG@5 scores. For the Fashion200K, FashionIQ and OKVQA datasets, we report Recall@10 scores, while for all other datasets, we report Recall@5 scores.

5 Experiments

5.1 Settings

Training Data Building on the findings from Section 3, we trained our model using a diverse dataset of 8 million instances spanning various retrieval modalities. For single-modal retrieval tasks, we utilized datasets including MSMARCO (Nguyen et al., 2016), NQ (Kwiatkowski et al., 2019), HotpotQA (Yang et al., 2018), TriviaQA (Joshi et al., 2017), SQuAD (Rajpurkar et al., 2016), FEVER (Thorne et al., 2018), and AllNLI for SimCSE (Gao et al., 2021), selecting a total of 1 million entries. From ImageNet (Deng et al., 2009), we extracted 1 million image-to-image training instances, designating images within the same class as positive samples and others as negative samples. For cross-modal retrieval tasks, we incorporated 2 million entries from the LAION (Schuhmann et al., 2022), mscoco (Lin et al., 2014), and Docmatix (Laurençon et al., 2024) datasets. Additionally, for fused-modal retrieval tasks, we included a total of 2 million instances: 1.1 million synthesized by us, and the remaining from the M-BEIR (Wei et al., 2024) training data.

Training Configuration We use Qwen2-VL (Wang et al., 2024b) model series as the backbone for our MLLM, conducting training on models with both 2 billion (2B) and 7 billion (7B) parameters. Our training utilizes Low-Rank Adaptation (LoRA) (Hu et al., 2022) with a rank of 8, a learning rate of 1e-4, and a temperature setting of 0.03. To manage the varying number of visual tokens required by Qwen2-VL for different image resolutions and maintain training efficiency, we limited the maximum number of visual tokens per image to 1,024.

For data with images, we set the maximum text length to 1,800 tokens, using a batch size of 128 for the 2B model and 32 for the 7B model. For text-only data, the maximum length was set to 512 tokens, with batch sizes of 512 for the 2B model and 128 for the 7B model. Each training sample included 8 negative examples. To conserve GPU memory, we employed gradient checkpointing (Chen et al., 2016) and trained the model using bfloat16 precision. All training was conducted on eight NVIDIA A100 GPUs, each with 80GB of memory.

Baselines We compare our method against four types of retrieval systems: (1) Previous representative UMR models, for example, VISTA (Zhou et al., 2024a) for text encoder based, and E5-V (Jiang et al., 2024a) for MLLM based; (2) Powerful multimodal representation (embedding) models, *i.e.*, One-Peace (Wang et al., 2023), which supports modalities beyond text and image and hence could also be tested on our UMRB; (3) Recent visual document retrieval models, namely DSE (Ma et al., 2024); and (4) the classic cross-modal pipeline, CLIP score-fusion, denoted as CLIP-SF, which provides top-tier cross-modal performance. We exclude comparisons with state-of-the-art text retrieval models as VISTA demonstrates comparable performance levels.

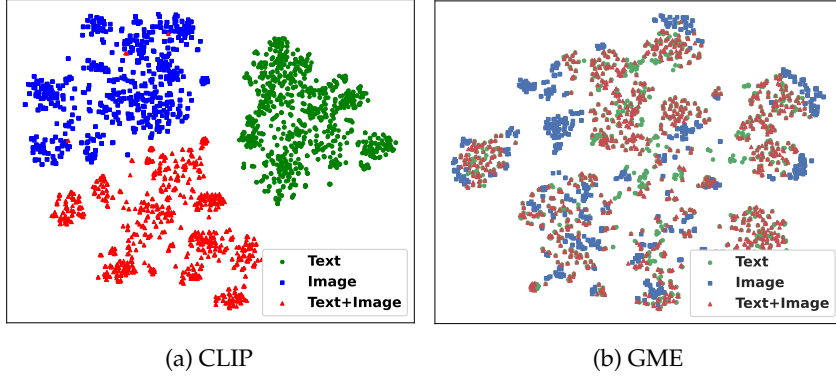


Figure 5: Distribution of different modal’s representation of CLIP (left) and ours GME-QWEN2-VL-2B (right).

5.2 Main Results

Table 3 presents the evaluation results of the baseline systems alongside our proposed GME. Scores are averaged across each sub-task and categorized by retrieval modality type: single-modal, cross-modal, and fused-modal. Additionally, the overall micro-average score on the UMRB is in the last column. First, focusing on the average scores, our smaller model, *i.e.*, GME-Qwen2-VL-2B, already outperforms the previous state-of-the-art UMR model (VISTA). The larger model, *i.e.*, GME-Qwen2-VL-7B, further enhances this performance, demonstrating the effectiveness of our approach in handling UMR tasks.

Second, our models outperform smaller methods such as VISTA (million-level parameters) and One-Peace (4B parameters). The larger MLLM baseline, E5-V (8B parameters), performs well in text-dominated tasks (e.g., $T \rightarrow T$) but falls short in other areas. This indicates that training with multimodal data is crucial for achieving superior performance in UMR tasks. Our training data provides a stronger foundation for future advancements.

Next, the cross-modal pipeline CLIP-SF outperforms UMR models like VISTA, E5-V, and One-Peace. For VISTA and E5-V, the performance gap is likely due to limitations in their text-modality bounds: VISTA is constrained by the text embedding space of its fixed backbone, and E5-V is limited by text-only training. One-Peace’s modality alignment-centered modeling may not be optimized for fused-modal content. In contrast, our models are specifically designed to handle fused-modal data, resulting in significantly better performance compared to the baselines. Although our training data includes several previously constructed fused-modal datasets, the contribution of our generated fused-modal training data will be discussed in §5.3.

Finally, we compare with the recent visual document retrieval model DSE, specialized for the $T \rightarrow VD$ task within the Cross-Modal group, which has approximately 4B parameters. Our models are competitive with or exceed the performance of this task-specific baseline, demonstrating the feasibility and promise of integrating visual document retrieval into a unified retriever framework.

5.3 Analyses

In this subsection, we conduct several analyses to further understand our methods and models.

Are the Produced Embeddings Modality Universal? Given our model’s impressive performance, we assess the quality of its embeddings. Specifically, we investigate whether the embeddings are modality-universal meaning that embeddings representing the same semantic content across different modalities are closely clustered in the embedding space, or if they remain in separate sub-spaces tailored for each modality-specific task. To probe this

Setting	Single	Cross	Fused	Average
w/ EVQA	45.13	60.21	49.32	51.55
w/ Gen _{Flux}	46.27	61.19	51.46	52.97
w/ Gen _{Google}	47.08	61.35	52.01	53.48

Table 4: Results of GME-Qwen2-VL-2B Trained with Various Datasets on UMRB-Partial.

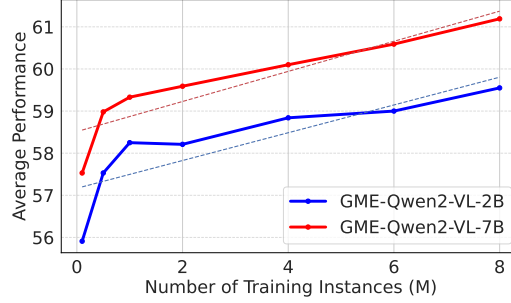


Figure 6: Average Performance of GME-Qwen2-VL-2B (Blue) and GME-Qwen2-VL-7B (Red) on UMRB-Partial, trained with varying numbers of training instances.

question, we sample 1000 instances from the EVQA dataset and visualize their embeddings of different modalities by t-SNE, as shown in Figure 5. We can observe that the CLIP’s embeddings are distinctly separated by modality, whereas the embeddings from our model are intermingled and organized semantically. This demonstrates that our model effectively generates modality-universal representations, enhancing its applicability across various UMR tasks.

Ablation Study on Synthetic Fused-Modal Data We propose an efficient data synthesis pipeline (§4.2) and generate large-scale fused-modal pairs to support model training. After witnessing the state-of-the-art performance of our model, it is natural to question the contribution of this synthetic data to the overall performance. To this end, we conduct an ablation study using three parallel training datasets, each comprising 100,000 pairs: original EVQA data, synthetic data with Google-retrieved images (Gen_{Google}), and synthetic data with FLUX-generated images (Gen_{Flux}). We train three models with identical parameters on these datasets and evaluate their performance on UMRB-Partial, with results shown in Table 4. Both synthetic datasets outperform the original EVQA data, indicating the high quality of our synthesized data. Although Google-retrieved images achieved marginally better performance than FLUX-generated images, the difference is minor and acceptable given the potential limitations of the Google Search API for rapid, large-scale dataset generation.

Training Scaling Law Our approach is primarily data-centric, constructing a diverse training dataset of approximately 8 million samples across various UMR settings (§5.1). Training on such a large-scale dataset demands significant computational resources and time, making arbitrary training decisions impractical. Therefore, we explored the training scaling law by examining how model performance evolves with increasing training steps. Due to the time-consuming nature of evaluating certain retrieval tasks, we assessed performance on our UMRB-Partial dataset for faster evaluation. Figure 6 illustrates the performance progression of our 2B and 7B models on UMRB-Partial during training. Both models exhibit linear performance improvements as training continues, suggesting that extended training could yield further benefits. However, due to time constraints, we halted current training. Future work will investigate longer training periods to enhance model performance further.

Ablation Study on Modeling We conduct an ablation study to investigate the effectiveness of different design choices of GME. We consider the following three aspects: (1) Fine-tuning strategy. Our final models are trained by LoRA with rank 8. We compare with other rank

Setting	Single	Cross	Fused	Average
Fine-tuning strategy				
LoRA r=8	48.09	78.39	51.88	59.45
LoRA r=16	47.86	78.63	51.42	59.30
LoRA r=32	47.85	78.55	50.48	58.96
LoRA r=64	47.65	78.61	51.09	59.11
Full training	43.16	75.79	49.28	56.07
Training data organization				
w/o hard-negative	47.55	78.01	50.95	58.83
w/o Instruction	46.82	78.10	49.09	58.00
Modeling				
w/ mean pooling	47.86	77.95	51.33	59.04
w/ bi-attention	46.55	76.78	49.54	57.62

Table 5: Results of the ablation study on Qwen2-VL-2B. All models were trained using 100,000 instances, consistent with the experimental setup described in Section 4.1.

values and full fine-tuning. The results in the first group of Table 5 show that LoRA with rank 8 yields the best performance. (2) Training data organization. We compare models trained without hard negative mining or retrieval instructions. The second group of Table 5 demonstrates that both modifications led to performance declines, indicating that these two techniques are essential for effective model training. (3) Modeling techniques. Our final models are in the casual attention mode and use the EOS token state as the embedding, hence we compare the performance of the model trained with mean pooling and the bi-directional attention mechanism. The last group of Table 5 shows that these alternative settings negatively impact performance.

6 Conclusion

In this work, we target the universal multimodal retrieval (UMR) problem. We begin by systematically categorizing current UMR tasks, proposing a comprehensive classification framework. Based on this, we explore ways to further improve MLLM-based UMR models, suggesting the GME model. The GME models are trained using contrastive learning loss on a diverse set of multimodal data settings, while also extending support for visual retrieval. Additionally, to overcome limitations in existing UMR evaluation benchmarks, we compiled a new comprehensive benchmark (i.e., UMRB) by integrating multiple data sources. This benchmark effectively balances existing UMR tasks with the increasingly important text and visual document retrieval tasks, enabling a more thorough assessment of UMR model performance. We evaluate existing UMR models and our proposed GME model on UMRB, finding that our model achieves state-of-the-art performance. We also conducted various analyses to validate the effectiveness of our methods and enhance our understanding of them.

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Appendix

A UMRB Details

Table 6 summarizes all UMRB tasks along with their statistics. Table 14 provides examples of different task types. Below is a brief description of each dataset included in the UMRB.

A.1 Single-Modal Tasks

WebQA (Chang et al., 2022) This dataset is derived from Wikipedia. In the T→T setup, both the query and candidate are text. The objective is to find a Wikipedia paragraph that answers the question. We have used 2,455 samples as the test set.

Nights (Fu et al., 2023) This dataset contains human judgments on the similarity of various image pairs, where both the query and candidate are images. The task is to identify an image that resembles the provided query image. We included 2,120 samples in our UMRB.

ArguAna, ClimateFEVER, CQADupstack, DBPedia, FEVER, FiQA2018, HotpotQA, MS-MARCO, NFCorpus, NQ, Quora, SCIDOCS, SciFact, Touche2020 and TRECCOVID For these datasets, we use the processed versions from BEIR (Thakur et al., 2021).

A.2 Cross-Modal Tasks

VisualNews (Liu et al., 2021a) This dataset focuses on the news domain and consists of pairs of news headlines and associated images. In UMRB, this dataset can be transformed into two tasks: retrieving the corresponding image based on the news headline (T→I) and retrieving the corresponding news headline based on the image (I→T). We utilized 19,995 and 20,000 samples to construct the test set.

Fashion200k (Han et al., 2017) This dataset includes pairs of images and product descriptions. In total, we have 1,719 instances for the task T→I and 4,889 instances for the task I→T for evaluation.

MSCOCO (Lin et al., 2014) This dataset is a well-known image caption dataset. Similar to VisualNews, it is converted into two tasks: “I→T”, which retrieves the caption given an image and “T→I”, which retrieves the image given a caption.

Name	Type	Categ.	Eval Samples	Candidates Nums	Eval Query avg. chars	Eval Candidate avg. chars	In partial
ArguAna	Single-Modal	T→T	10,080	1,406	192.98	166.80	True
Climate-FEVER	Single-Modal	T→T	1,535	5,416,593	20.13	84.76	False
CQADupStack	Single-Modal	T→T	13,145	457,199	8.59	129.09	False
DBPedia	Single-Modal	T→T	400	4,635,922	5.39	49.68	False
FEVER	Single-Modal	T→T	6,666	5,416,568	8.13	84.76	False
FiQA2018	Single-Modal	T→T	648	57,638	10.77	132.32	False
HotpotQA	Single-Modal	T→T	7,405	5,233,329	17.61	46.30	False
MSMARCO	Single-Modal	T→T	6,980	8,841,823	5.96	55.98	False
NFCorpus	Single-Modal	T→T	323	3,633	3.30	232.26	True
NQ	Single-Modal	T→T	3,452	2,681,468	9.16	78.88	False
Quora	Single-Modal	T→T	10,000	522,931	9.53	11.44	True
SCIDOCS	Single-Modal	T→T	1,000	25,657	9.38	176.19	True
SciFact	Single-Modal	T→T	300	5,183	12.37	213.63	False
Touche2020	Single-Modal	T→T	49	382,545	6.55	292.37	False
TRECCOVID	Single-Modal	T→T	50	171,332	10.60	160.77	True
WebQA	Single-Modal	T→T	2,455	544,457	18.58	37.67	False
Nights	Single-Modal	I→I	2,120	40,038	-	-	True
VisualNews	Cross-Modal	T→I	19,995	542,246	18.78	-	False
Fashion200k	Cross-Modal	T→I	1,719	201,824	4.89	-	False
MSCOCO	Cross-Modal	T→I	24,809	5,000	10.43	-	True
Flickr30k	Cross-Modal	T→I	5,000	1,000	12.33	-	True
TAT-DQA	Cross-Modal	T→VD	1,646	277	12.44	-	False
ArxivQA	Cross-Modal	T→VD	500	500	17.12	-	False
DocVQA	Cross-Modal	T→VD	451	500	8.23	-	True
InfoVQA	Cross-Modal	T→VD	494	500	11.29	-	False
Shift Project	Cross-Modal	T→VD	100	1,000	16.01	-	True
Artificial Intelligence	Cross-Modal	T→VD	100	968	12.3	-	False
Government Reports	Cross-Modal	T→VD	100	972	12.62	-	False
Healthcare Industry	Cross-Modal	T→VD	100	965	12.56	-	False
Energy	Cross-Modal	T→VD	100	977	13.49	-	False
TabFQuad	Cross-Modal	T→VD	280	70	16.49	-	False
VisualNews	Cross-Modal	I→T	20,000	537,568	-	18.53	False
Fashion200k	Cross-Modal	I→T	4,889	61,707	-	4.95	False
MSCOCO	Cross-Modal	I→T	5,000	24,809	-	10.43	True
Flickr30k	Cross-Modal	I→T	1,000	5,000	-	12.33	True
WebQA	Fused-Modal	T→IT	2,511	403,196	16.43	12.83	False
EDIS	Fused-Modal	T→IT	3,241	1,047,067	20.07	15.53	False
OVEN	Fused-Modal	IT→T	50,004	676,667	6.52	82.13	False
INFOSEEK	Fused-Modal	IT→T	11,323	611,651	8.76	91.49	False
ReMuQ	Fused-Modal	IT→T	3,609	138,794	13.82	34.26	True
OKVQA	Fused-Modal	IT→T	5,046	114,516	8.09	102.55	True
LLaVA	Fused-Modal	IT→T	5,120	5,994	10.70	90.65	True
FashionIQ	Fused-Modal	IT→I	6,003	74,381	11.70	-	True
CIRR	Fused-Modal	IT→I	4,170	21,551	11.01	-	True
OVEN	Fused-Modal	IT→IT	14,741	335,135	5.91	94.76	True
EVQA	Fused-Modal	IT→IT	3,743	68,313	9.38	211.12	False
INFOSEEK	Fused-Modal	IT→IT	17,593	481,782	7.94	96.00	False

Table 6: Tasks in UMRB. We counted the number of datasets under each task type and the number of evaluation instances in the dataset, the size of the candidate set, and the average length of the text.

Flickr30k(Plummer et al., 2015) This dataset consists of images paired with detailed textual descriptions. We have a total of 1,000 instances for the I→T task and 5,000 instances for the T→I task available for evaluation.

TAT-DQA, ArxivQA, DocVQA, InfoVQA, Shift Project, Artificial Intelligence, Government Reports, Healthcare Industry, Energy, TabFQuad These datasets constitute the retrieval task of T→VD. Their queries are standard questions, and the candidates are document screenshots. For these datasets, we used the processed versions from ViDoRe (Faysse et al., 2024b).

A.3 Fused-Modal Tasks

WebQA (Chang et al., 2022) Similar to WebQA in the Single-Modal setting, this dataset is also derived from Wikipedia, but in the T→IT setup, the candidates consist of images

and text. The task is to find a Wikipedia paragraph with accompanying text and images to answer a specific question. There are 2,511 samples in the evaluation set.

EDIS (Liu et al., 2023b) This dataset involves the cross-modal image search within the news domain. The queries are texts containing entities and events, with candidates consisting of news images and their accompanying headlines. The task requires the model to comprehend both entities and events from the text queries and retrieve the corresponding image and headline.

OVEN (Hu et al., 2023) The dataset is sourced from Wikipedia, where a query consists of an image and a question related to the image. The candidates are the Wikipedia title along with the first 100 tokens of its summary. If the associated Wikipedia content includes images, it constitutes an IT→IT task; otherwise, it forms an IT→T task. In the evaluation, we have 14,741 samples for the IT→IT task and 50,004 samples for the IT→T task.

INFOSEEK (Chen et al., 2023) This dataset is similar to OVEN, with queries consisting of images alongside text questions. The candidates are Wikipedia snippets of 100 tokens containing the exact answers. This dataset also encompasses two tasks: for the IT→IT and IT→T tasks, we used 17,593 and 11,323 samples, respectively.

ReMuQ (Luo et al., 2023) The dataset is designed for a multimodal-query retriever, augmenting the WebQA questions by adding images to create new multimodal queries along with a large text corpus. For evaluation, we used 3,609 instances from this dataset.

OKVQA (Marino et al., 2019) This dataset includes visual questions that require external knowledge to answer. It is structured as an IT→T retrieval task, where queries consist of visual questions containing images and text, with candidates being external knowledge sources that can assist in answering the questions.

LLaVA (Lin et al., 2024) This dataset contains high-quality conversations about an image generated by GPT-3.5, involving exchanges between a human and an AI assistant. The queries comprise questions and instructions sent by humans to the AI assistant, which include both images and text, while the candidates are the AI assistant’s replies. We utilized 5,120 samples from this dataset in the UMRB evaluation.

FashionIQ (Wu et al., 2021) This dataset features images of fashion products along with crowd-sourced descriptions that highlight the differences between these products. Each query consists of an image and a modification sentence that describes changes to the given image, with the retrieval target being the specified image. In the UMRB evaluation, we used 6,003 samples from this dataset.

CIRR (Liu et al., 2021b) Similar to FashionIQ, CIRR can also be used for composed image retrieval. It involves pairs of real-life reference and target images in each test case, along with a modification sentence detailing the differences between the two images. For the UMRB evaluation, we utilized 4,170 samples from this dataset.

EVQA (Mensink et al., 2023) This dataset is akin to INFOSEEK, with the key distinction being that the retrieval target of EVQA is a complete Wikipedia paragraph with a maximum length of several thousand tokens. We used 3,743 samples for evaluation, eliminating multi-hop issues present in the original test set. We selected Wikipedia paragraphs from the original dataset as candidates and supplemented them with images. Images native to each paragraph were included when available; otherwise, the first image from the article was utilized due to its typically representative nature.

A.4 UMRB-Partial

The full UMRB dataset consists of 47 subtasks, approximately 200,000 evaluation instances, and 40 million candidates, resulting in a significant overhead when testing the model. Dur-

ing our experiments with the GME-7B model, a full evaluation required approximately 400 A100*80G GPU hours. To facilitate development and verification, we created a smaller benchmark by condensing the complete UMRB, which we refer to as **UMRB-Partial**. Column 8 of Table 2 indicates whether a dataset is included in **UMRB-Partial**. Testing the GME-7B model on UMRB-Partial reduced the evaluation time from 400 A100*80G GPU hours to 80 A100*80G GPU hours.

B Results Details

In this section, we present the detailed scores achieved by our GME and the baseline models on various tasks. Additionally, we provide results from other benchmarks, including BEIR, M-BEIR, and ViDoRe.

B.1 Detailed Results on UMRB

Table 7 presents the detailed evaluation results of the baseline systems alongside our GME on UMRB tasks. First, focusing on the average scores, our smaller model, *i.e.*, GME-Qwen2-VL-2B, already outperforms the previous state-of-the-art UMR model (VISTA). The larger model, *i.e.*, GME-Qwen2-VL-7B, further enhances this performance. In addition, focusing on specific scores on different datasets, our GME achieves state-of-the-art performance on each dataset except the Nights dataset. VISTA and CLIP-SF scored highly on the Nights dataset, likely due to their use of independent image and text encoders for cross-modal retrieval. In the I→I task, these models relied solely on the image encoder for encoding without cross-modal alignment, which may explain their superior performance on the Nights dataset.

B.2 Detailed Results on UMRB-Partial

Figure 3 illustrates our exploration of the training data, as discussed in Section 4.2, with specific results presented in Table 8. This table details the scores of our models trained on six data types: T→T, I→I, T→VD, T→I, IT→IT, and Mix across various tasks. We find that the model trained on mixed data performs the best.

B.3 Detailed Results on BEIR

BEIR is a heterogeneous benchmark containing diverse text IR tasks. We utilize BEIR to compare the performance of our GME with other text embedders on T→T tasks. Table 9 presents the detailed evaluation nDCG@10 scores for pure text embedders and multimodal embedders on T→T tasks. Except for our GME, other multimodal embedders do not match the performance of pure text embedders on text retrieval tasks, including those like E5-V that are fine-tuned exclusively on text data.

Naturally, pure text embedding models of the same model size still outperform multimodal embedding models in pure text retrieval tasks. For example, the score of the gte-Qwen2-7B-instruct model is 60.25, while the GME-Qwen2-VL-7B model, with the same model scale, scores 55.63. Although both models share the same text LLM, incorporating or extending multimodal capabilities leads to additional compromises in pure text performance. Minimizing this kind of loss remains an important research question.

B.4 Detailed Results on M-BEIR

M-BEIR, the Multimodal Benchmark for IR, serves as a comprehensive large-scale retrieval benchmark designed to evaluate multimodal retrieval models. As shown in Table 10, we report Recall@10 scores for the Fashion200K and FashionIQ datasets, while Recall@5 scores are provided for all other datasets. In M-BEIR, our GME continues to demonstrate state-of-the-art performance, underscoring the effectiveness of our approach.

Type	Task	Dataset	VISTA	CLIP-SF	One-Peace	DSE	E5-V	GME-2B	GME-7B
Single-Modal	T→T	ArguAna	63.61	52.45	32.93	53.46	54.28	61.52	64.60
		Climate-FEVER	31.17	20.00	20.27	19.79	21.64	42.30	45.38
		CQADupStack	42.35	30.61	41.32	36.51	41.69	38.13	41.66
		DBPedia	40.77	26.37	32.43	40.75	38.78	46.31	50.78
		FEVER	86.29	50.58	51.91	80.12	78.99	92.60	94.27
		FiQA2018	40.65	22.14	36.79	36.2	45.41	45.30	57.14
		HotpotQA	72.6	41.33	46.51	70.79	60.88	72.93	79.21
		MSMARCO	41.35	22.15	36.55	37.73	41.23	40.88	42.38
		NFCorpus	37.39	27.05	31.6	32.82	36.97	37.20	38.40
		NQ	54.15	25.45	42.87	52.97	51.58	60.01	67.04
		Quora	88.90	81.63	87.46	85.84	87.6	87.24	88.05
		SCIDOCS	21.73	14.75	21.64	15.66	22.36	23.17	27.38
		SciFact	74.04	55.98	64.51	68.97	72.75	63.82	62.31
		Touche2020	25.7	17.47	16.90	14.50	21.61	29.06	23.26
		TRECCOVID	77.90	63.61	69.28	52.98	72.85	59.24	52.60
	WebQA	83.80	84.44	63.67	83.95	89.94	95.19	95.85	
I→I	Nights	24.43	31.42	31.27	27.36	27.92	29.86	31.89	
Cross-Modal	T→I	VisualNews	5.77	42.80	48.95	14.12	29.46	38.85	46.54
		Fashion200k	3.08	18.38	32.34	3.08	3.78	25.83	31.82
		MSCOCO	47.97	80.75	71.45	74.62	52.38	71.82	75.14
		Flickr30k	74.68	94.28	92.78	94.42	77.38	92.92	91.92
	T→VD	TAT-DQA	2.05	5.49	14.44	49.01	9.08	71.05	76.12
		ArxivQA	10.30	24.10	43.94	78.17	41.16	83.91	87.58
		DocVQA	8.01	11.80	23.48	45.83	24.37	54.57	56.63
		InfoVQA	30.02	48.78	59.97	82.06	49.5	91.11	92.39
		Shift Project	3.26	6.06	17.02	69.84	13.16	94.29	97.26
		Artificial Intelligence	7.34	28.64	45.41	96.88	46.18	99.02	99.63
		Government Reports	6.90	34.67	55.98	92.04	53.05	98.89	99.63
		Energy	11.05	27.19	53.21	92.62	56.77	93.15	95.89
		TabFQuad	13.08	21.53	57.05	79.29	58.22	94.61	94.58
	Healthcare Industry	9.39	32.64	59.55	96.35	59.61	98.89	99.63	
	I→T	VisualNews	2.79	42.67	47.27	8.74	29.54	38.32	45.54
		Fashion200k	4.72	18.10	30.89	3.91	4.62	27.57	34.20
MSCOCO		48.92	91.94	85.6	82.06	86.4	84.12	84.90	
Flickr30k		68.50	99.11	98.60	97.11	89.62	97.70	98.70	
Fused-Modal	T→IT	WebQA	54.84	78.42	32.42	66.99	49.62	83.15	84.59
		EDIS	36.78	54.09	53.01	41.26	49.62	70.32	77.29
	IT→T	OVEN	22.32	45.98	23.69	0.38	14.4	58.17	63.41
		INFOSEEK	18.53	27.58	20.05	3.06	12.69	39.06	43.14
		ReMuQ	76.20	83.71	26.41	94.60	52.15	98.75	98.12
		OKVQA	17.14	17.44	9.67	13.28	16.71	28.38	27.03
		LLaVA	72.81	91.91	51.64	53.18	77.48	98.54	99.18
	IT→I	FashionIQ	3.28	24.54	2.93	9.81	3.73	27.50	31.43
		CIRR	14.65	45.25	10.53	36.52	13.19	46.83	53.69
	IT→IT	OVEN	27.77	68.83	30.56	0.39	54.46	75.98	80.30
		EVQA	28.75	40.08	16.64	15.34	26.39	79.21	81.75
		INFOSEEK	22.27	49.05	23.32	5.96	39.69	44.21	58.80
Avg.			37.32	43.66	42.01	50.04	42.52	64.45	67.44

Table 7: The detailed results of the baselines and our GME on UMRB. Following previous works Thakur et al. (2021); Wei et al. (2024); Faysse et al. (2024b), we present NDCG@10 scores for T→T tasks, excluding the WebQA dataset. For T→VD tasks, we provide NDCG@5 scores. For the Fashion200K, FashionIQ and OKVQA datasets, we report Recall@10 scores, while for all other datasets, we report Recall@5 scores.

Type	Task	Dataset	T→T	I→I	T→VD	T→I	IT→IT	Mix
Single-Modal	T→T	Arguan	56.25	43.51	56.73	33.53	53.22	56.22
		NFCorpus	35.23	28.89	33.23	33.18	30.48	35.76
		Quora	87.82	74.37	86.32	86.43	85.2	87.4
		SCIDOCs	19.07	11.82	17.51	17.2	16.93	19.88
		TRECCOVID	75.57	47.89	50.89	72.37	58.92	76.38
	I→I	Nights	27.97	28.11	24.9	28.53	26.04	30.85
Cross-Modal	T→I	MSCOCO	59.7	59.41	63.67	76.91	44.97	75.3
		Flickr30k	83.92	65.52	87.32	93.18	74.52	93.06
	T→VD	DocVQA	35.8	24.24	48.38	40.58	28.05	45.62
		Shift Project	57.86	45.47	77.08	50.36	53.12	74.84
	I→T	MSCOCO	74.72	63.82	80.46	84.64	70.48	84.24
		Flickr30k	94.1	82.5	96.3	97.2	90.1	97.5
Fused-Modal	IT→T	LLaVA	92.75	89.05	86.02	89.24	88.73	95.02
		ReMuQ	89.61	85.47	76.45	85.12	86.73	89.75
		OKVQA	24.55	16.6	15.78	16.92	18.57	20.23
	IT→I	FashionIQ	5.53	4.2	5.43	8.86	11.08	11.89
		CIRR	17.24	15.04	15.42	17.5	25.71	29.86
	IT→IT	OVEN	59.81	38.42	57.31	56.69	65.08	63.04
Avg.			55.42	45.80	54.50	54.91	51.55	60.38

Table 8: Performance of models trained on different data types on UMR-partial. We present NDCG@10 scores for T→T tasks. For T→VD tasks, we provide NDCG@5 scores. For the FashionIQ dataset, we report Recall@10 scores, while for all other datasets, we report Recall@5 scores.

BEIR	Avg.	Argu- Ana	Cli- mate- Fever	CQA- Dup- Stack	DB- Pedia	Fever	FiQA	Hotpot- QA	MS MAR- CO	NF- Corpus	NQ	Quora	Sci- docs	Sci- fact	Touche- 2020	Trec- Covid
Text Embedder																
gte-Qwen2-7B-instruct	60.25	64.27	45.88	46.43	52.42	95.11	62.03	73.08	45.98	40.6	67	90.09	28.91	79.06	30.57	82.26
NV-Embed-v1	59.36	68.2	34.72	50.51	48.29	87.77	63.1	79.92	46.49	38.04	71.22	89.21	20.19	78.43	28.38	85.88
gte-Qwen2-1.5B-instruct	58.29	69.72	42.91	44.76	48.69	91.57	54.7	68.95	43.36	39.34	64	89.64	24.98	78.44	27.89	85.38
voyage-large-2-instruct	58.28	64.06	32.65	46.6	46.03	91.47	59.76	70.86	40.6	40.32	65.92	87.4	24.32	79.99	39.16	85.07
neural-embedding-v1	58.12	67.21	32.3	49.11	48.05	89.46	58.94	78.87	42	42.6	68.36	89.02	27.69	78.82	24.06	75.33
GriffLM-7B	57.41	63.24	30.91	49.42	46.6	82.74	59.95	79.4	41.96	40.89	70.3	89.47	24.41	79.17	27.93	74.8
e5-mistral-7b-instruct	56.89	61.88	38.35	42.97	48.89	87.84	56.59	75.72	43.06	38.62	63.53	89.61	16.3	76.41	26.39	87.25
google-gecko	55.7	62.18	33.21	48.89	47.12	86.96	59.24	71.33	32.58	40.33	61.28	88.18	20.34	75.42	25.86	82.62
text-embedding-3-large	55.44	58.05	30.27	47.54	44.76	87.94	55	71.58	40.24	42.07	61.27	89.05	23.11	77.77	23.35	79.56
gte-en-large-v1.5	57.91	72.11	48.36	42.16	46.3	93.81	63.23	68.18	42.93	36.95	56.08	89.67	26.35	82.43	22.55	77.49
gte-en-base-v1.5	54.09	63.49	40.36	39.52	39.9	94.81	48.65	67.75	42.62	35.88	52.96	88.42	21.92	76.77	25.22	73.13
Multimodal Embedder																
VISTA	53.24	63.61	31.17	42.35	40.77	86.29	40.65	72.6	41.35	37.39	54.15	88.9	21.73	74.04	25.7	77.9
CLIP-SF	36.77	52.45	20	30.61	26.37	50.58	22.14	41.33	22.15	27.05	25.45	81.63	14.75	55.98	17.47	63.60
One-Peace	42.19	32.93	20.27	41.32	32.43	51.91	36.79	46.51	36.55	31.6	42.87	87.46	21.64	64.51	16.9	69.28
DSE	46.60	53.46	19.79	36.51	40.75	80.12	36.2	70.79	37.73	32.82	52.97	85.84	15.66	68.97	14.50	52.98
E5-V	49.91	54.28	21.64	41.69	38.78	78.99	45.41	60.88	41.23	36.97	51.58	87.6	22.36	72.75	21.61	72.85
GME-Qwen2VL-2B	53.31	61.52	42.30	38.13	46.31	92.60	45.30	72.93	40.88	37.20	60.01	87.24	23.17	63.82	29.06	59.24
GME-Qwen2VL-7B	55.63	64.60	45.38	41.66	50.78	94.27	57.14	79.21	42.38	38.40	67.04	88.05	27.38	62.31	23.26	52.60

Table 9: BEIR benchmark (Thakur et al., 2021) nDCG@10 scores. We include top models from MTEB Retrieval English leaderboard.

MBEIR	Avg.	$q_i \rightarrow c_i$			$q_i \rightarrow c_i$		$q_i \rightarrow (c_i, c_i)$		$q_i \rightarrow c_i$			$(q_i, q_i) \rightarrow c_i$		$(q_i, q_i) \rightarrow c_i$		$(q_i, q_i) \rightarrow (c_i, c_i)$	
		Visual-News	MS-COCO	Fashion-200K	Web-QA	EDIS	Web-QA	Visual-News	MS-COCO	Fashion-200K	NIGHTS	OVEN	Info-Seek	Fashion-IQ	CIRR	OVEN	Info-Seek
CLIP	32.5	43.3	61.1	6.6	36.2	43.3	45.1	41.3	79.0	7.7	26.1	24.2	20.5	7.0	13.2	38.8	26.4
SigLIP	37.2	30.1	75.7	36.5	39.8	27.0	43.5	30.8	88.2	34.2	28.9	29.7	25.1	14.4	22.7	41.7	27.4
BLIP	26.8	16.4	74.4	15.9	44.9	26.8	20.3	17.2	83.2	19.9	27.4	16.1	10.2	2.3	10.6	27.4	16.6
BLIP2	24.8	16.7	63.8	14.0	38.6	26.9	24.5	15.0	80.0	14.2	25.4	12.2	5.5	4.4	11.8	27.3	15.8
VISTA	26.37	5.77	47.97	3.08	83.80	36.78	54.84	2.79	48.92	4.72	24.43	22.32	18.53	3.28	14.65	27.77	22.27
CLIP-SF	50.26	42.80	80.75	18.38	84.44	54.09	78.42	42.67	91.94	18.10	31.42	45.98	27.58	24.53	45.25	68.83	49.05
One-Peace	38.00	48.95	71.45	32.34	63.67	53.01	32.42	47.27	85.60	30.89	31.27	23.69	20.05	2.93	10.53	30.56	23.32
DSE	28.89	14.12	74.62	3.08	83.95	41.26	66.99	8.74	82.06	3.91	27.36	0.38	3.06	9.81	36.52	0.39	5.96
E5-V	35.09	29.46	52.38	3.78	89.94	49.62	49.62	29.54	86.40	4.62	27.92	14.40	12.69	3.73	13.19	54.46	39.69
GME-Qwen2-VL-2B	53.55	38.85	71.82	25.83	95.19	70.32	83.15	38.32	84.12	27.57	29.86	58.17	39.06	27.50	46.83	75.98	44.21
GME-Qwen2-VL-7B	58.66	46.54	75.14	31.82	95.85	77.29	84.59	45.54	84.90	34.20	31.89	63.41	43.14	31.43	53.69	80.30	58.80

Table 10: MBEIR benchmark Wei et al. (2024). For the Fashion200K and FashionIQ datasets, we report Recall@10 scores, while for all other datasets, we report Recall@5 scores.

	ArxivQ	DocQ	InfoQ	TabF	TATQ	Shift	AI	Energy	Gov.	Health.	Avg.
BM25Text + Captioning	40.1	38.4	70.0	35.4	61.5	60.9	88.0	84.7	82.7	89.2	65.1
BGE-M3Text + Captioning	35.7	32.9	71.9	69.1	43.8	73.1	88.8	83.3	80.4	91.3	67.0
Jina-CLIP	25.4	11.9	35.5	20.2	3.3	3.8	15.2	19.7	21.4	20.8	17.7
Nomic-vision	17.1	10.7	30.1	16.3	2.7	1.1	12.9	10.9	11.4	15.7	12.9
SigLIP (Vanilla)	43.2	30.3	64.1	58.1	26.2	18.7	62.5	65.7	66.1	79.1	51.4
ColPali	79.1	54.4	81.8	83.9	65.8	73.2	96.2	91.0	92.7	94.4	81.3
VISTA	10.3	8.01	30.02	13.08	2.05	3.26	7.14	11.05	6.9	9.39	10.12
CLIP-SF	24.1	11.8	48.78	21.53	5.49	6.06	28.64	27.19	34.67	32.64	24.09
One-Peace	43.94	23.48	59.97	57.05	13.44	17.02	45.41	53.21	55.98	59.5	42.9
DSE	78.17	45.83	82.06	79.29	49.01	69.84	96.89	92.62	92.04	96.35	78.21
E5-V	41.16	24.37	49.5	58.22	9.08	13.26	46.18	57.77	53.05	59.61	41.22
GME-Qwen2-VL-2B	83.91	54.57	91.11	94.61	71.05	94.29	99.02	93.15	98.89	98.89	87.84
GME-Qwen2-VL-7B	87.58	56.63	92.39	94.58	76.12	97.26	99.63	95.89	99.63	99.63	89.92

Table 11: Comprehensive evaluation of baseline models and our GME on ViDoRe. Results are presented using NDCG@5 metrics.

B.5 Detailed Results on ViDoRe

ViDoRe represents the Visual Document Retrieval Benchmark, encompassing various page-level screenshot retrieval tasks. This benchmark includes the $T \rightarrow VD$ tasks within our UMRB. Table 11 presents the detailed nDCG@5 scores for our GME and other models. Our smaller model, *i.e.*, GME-Qwen2-VL-2B, surpasses the previous state-of-the-art model (ColPali), which was exclusively trained on this dataset for this specific task. The larger model, *i.e.*, GME-Qwen2-VL-7B, further improves upon this performance.

C Experiment Details

C.1 Training Details

Our GME models (both 2B and 7B) are initialized using the Qwen2-VL (Wang et al., 2024b) model series. We employ the transformers library for training in BF16 precision. The training utilizes Low-Rank Adaptation (LoRA) (Hu et al., 2022) with a rank of 8. We apply a decoupled AdamW optimizer with a learning rate and a weight decay of $1e-4$. Additional hyperparameters are detailed in Table 12.

In our contrastive learning approach, we develop dense multimodal representation models (embedders) that utilize the [EOS] hidden state as the embedding of the input. The temperature for contrastive learning is set to 0.03. For each query, we include one positive candidate along with eight hard negative candidates.

C.2 Instructions

The complete UMRB consists of 47 tasks, each with distinct retrieval candidates and varying domains. Even within the same dataset, retrieval candidates can differ based on task types. For example, the WebQA dataset aims to retrieve textual candidates for $T \rightarrow T$ tasks, which is different from retrieving a combination of image and text candidates for $T \rightarrow IT$ tasks.

Hyper-param	GME-Qwen2-VL-2B	GME-Qwen2-VL-7B
Number of Params	2B	8.2B
Number of Layers	28	28
Hidden Size	1536	3584
FFN Inner Size		3072
Number of Attention Heads	12	28
Vision Depth		32
Vision Embed_dim		1280
Vision Patch_size		14
Temperature		0.03
Learning Rate Decay		Linear
Adam ϵ		1e-4
Adam β_1		0.9
Adam β_2		0.98
Gradient Clipping		0.0
Precision	PyTorch BF16 AMP	
Max Length	1800	1800
Batch Size	128	32
Warm-up Ratio		0.06

Table 12: GME training hyper-parameters.

We have designed specific instructions tailored for each task to guide the model in effectively completing the retrieval process. The detailed instructions are provided in Table 13.

D Fused-Modal Data Synthesis Details

We utilize doc2query to synthesize data. However, our goal is to generate fused-modal candidate-to-query relevance data rather than single-modality, text-based relevance pairs.

D.1 Prompts

Step 1: In the first step of data synthesis, we prompt the large language model (LLM) to generate a natural question and answer based on a selected passage. The specific prompt is illustrated in Figure 7. This process leverages in-context learning (ICL) to guide the LLM in producing outputs that align with our requirements.

Step 2: In step 2, we provide the LLM with the passage and the natural question generated in step 1. The LLM is then prompted to extract the main entity from the question and refactor the question accordingly. Figure 8 presents the prompt used in this step. In subsequent steps, the extracted entity will be replaced by the corresponding image, which, when combined with the reconstructed question, will form a fused-modal query.

Step 3: In step 3, we replace the entity with an image, which can be sourced in two ways. The first method involves prompting the LLM to generate a caption for the entity based on the provided entity and passage, after which the caption is fed into FLUX to generate images. The second method retrieves the entity by utilizing the Google Image Retrieval API. Figure 9 illustrates the caption generation prompt for this step.

D.2 Filter

Two filtering methods are implemented to ensure the quality of the synthesized data. First, a text retrieval model is utilized to evaluate unreconstructed queries and their corresponding passages. We follow the framework of Promptagator (Dai et al., 2023); a query is deemed unqualified if the passage that generated it does not appear within the top 20 search results. Second, for images obtained through the Google Image Search API, we employ the CLIP model to assess image-caption relevance. Images with a relevance score below 0.2 are filtered out.

Task	Dataset	Query Instruction
T→T	ArguAna	Given a claim, find documents that refute the claim.
	Climate-FEVER	Given a claim about climate change, retrieve documents that support or refute the claim.
	CQADupStack	Given a question, retrieve detailed question descriptions from Stackexchange that are duplicates to the given question.
	DBPedia	Given a query, retrieve relevant entity descriptions from DBPedia.
	FEVER	Given a claim, retrieve documents that support or refute the claim.
	FiQA2018	Given a financial question, retrieve user replies that best answer the question.
	HotpotQA	Given a multi-hop question, retrieve documents that can help answer the question.
	MSMARCO	Given a web search query, retrieve relevant passages that answer the query.
	NFCorpus	Given a question, retrieve relevant documents that best answer the question.
	NQ	Given a question, retrieve Wikipedia passages that answer the question.
	Quora	Given a question, retrieve questions that are semantically equivalent to the given question.
	SCIDocs	Given a scientific paper title, retrieve paper abstracts that are cited by the given paper.
	SciFact	Given a scientific claim, retrieve documents that support or refute the claim.
	Touche2020	Given a question, retrieve detailed and persuasive arguments that answer the question.
	TRECCOVID	Given a query on COVID-19, retrieve documents that answer the query.
	WebQA	Retrieve passages from Wikipedia that provide answers to the following question.
I→I	Nights	Find a day-to-day image that looks similar to the provided image.
T→I	VisualNews	Identify the news-related image in line with the described event.
	Fashion200k	Based on the following fashion description, retrieve the best matching image.
	MSCOCO	Identify the image showcasing the described everyday scene.
	Flickr30k	Find an image that matches the given caption.
T→VD	TAT-DQA ArxivQA DocVQA InfoVQA Shift Project Artificial Intelligence Government Reports Healthcare Industry Energy TabFQuad	Find a screenshot that relevant to the user's question.
I→T	VisualNews	Find a caption for the news in the given photo.
	Fashion200k	Find a product description for the fashion item in the image.
	MSCOCO	Find an image caption describing the following everyday image.
	Flickr30k	Find an image caption describing the following image.
T→IT	WebQA	Find a Wikipedia image that answers this question.
	EDIS	Identify the news photo for the given caption.
IT→T	OVEN INFOSEEK	Retrieve a Wikipedia paragraph that provides an answer to the given query about the image.
	ReMuQ	Retrieve a fact-based paragraph that provides an answer to the given query about the image.
	OKVQA	Retrieve documents that provide an answer to the question alongside the image.
	LLaVA	Provide a specific description of the image along with the following question.
IT→I	FashionIQ	Find a fashion image that aligns with the reference image and style note.
	CIRR	Retrieve a day-to-day image that aligns with the modification instructions of the provided image.
IT→IT	OVEN INFOSEEK	Retrieve a Wikipedia image-description pair that provides evidence for the question of this image.
	EVQA	Obtain illustrated documents that correspond to the inquiry alongside the provided image.

Table 13: The instructions for different tasks, we only use the instructions for query encoding

```
>> SYSTEM
You are a helpful assistant.

>> USER
Based on the provided **Passage**, generate a query and an answer.
The result should be directly returned in JSON format. Do not
generate any irrelevant information beyond the required output.
Example1:
**Passage**:
<passage>
**Output**:
{"query": "Is Heracleum mantegazzianum poisonous?", "answer": "yes"}

**Passage**:
<passage>
**Output**:
```

Figure 7: Fused-Modal Data Synthesis Step 1 Prompt.

```
>> SYSTEM
You are a helpful assistant.

>> USER
Extract the entities corresponding to the **Query** and **Passage**, and
replace the entities in the Query with general references such as "this
person, this building, this animal, this river, this bridge," etc. The
result should be returned in JSON format.
Example1:
**Query**:
Is Heracleum mantegazzianum poisonous?
**Passage**:
<passage>
**Output**:
{"entity": "Heracleum mantegazzianum", "query": "Is this plant poisonous?"}

**Query**:
<Query>
**Passage**:
<passage>
**Output**:
```

Figure 8: Fused-Modal Data Synthesis Step 2 Prompt.

Why is the threshold score set to 0.2? The relevance scores of all images searched via Google and the corresponding captions we have collected are presented in Figure 10. We select the median score of 0.2 to ensure image quality while also ensuring that most text queries have sufficient images to pair with.

D.3 Examples of synthetic data

Table 15 illustrates passages from 15 domains and the fused modal queries generated by applying the synthesis flow. “FLUX image” refers to images generated by the Vincennes diagram model FLUX.1-dev, whereas “Google image” indicates images from Google Image retrieval.

E Limitations

In this work, we present a benchmark for training and testing Universal Multimodal Retrieval (UMR). To better accomplish this task, we explore strategies for adapting Multimodal Large Language Models (MLLMs) into UMR models, presenting GME, a powerful embedding model capable of retrieving candidates across different modalities. However, this work has its limitations, which are outlined below:

1. Single Image Limit In MLLMs, one image is converted into a very large number of visual tokens. In Qwen2-VL, we limit the number of visual tokens to 1024. Due to model training efficiency and a lack of relevant data, our queries and candidates in UMRB only retain a single image. Thus, performance on interleaved data (where multiple images and texts are mixed together) cannot be assessed.

2. Single Language Limit Although the backbone of our model, Qwen2-VL, supports multiple languages, we only utilized a single language, English, during the training and

Type	Task	Query Text	Query Image	Target Text	Target Image
Single-Modal	T→T	where is whitmarsh island?	-	Whitemarsh Island, Georgia Whitemarsh Island, Georgia. Whitemarsh Island (pronounced WIT-marsh) is a census-designated place (CDP) in Chatham County, Georgia, United States. The population was 6,792 at the 2010 census. It is part of the Savannah Metropolitan Statistical Area. The communities of Whitemarsh Island are a relatively affluent suburb of Savannah.	-
	I→I	-		-	
Cross-Modal	T→I	Multicolor boutique amy black leather look biker jacket.	-	-	
	T→VD	Based on the graph, what is the impact of correcting for fspec not equal to 1 on the surface density trend?	-	-	
	I→T		-	Indian National Congress Vice President Rahul Gandhi addresses the special plenary session of Confederation of Indian Industr in New Delhi on April 4 2013.	-
Fused-Modal	T→IT	Does a Minnetonka Rhododendron flower have petals in a cup shape?	-	2020-05-08 15 17 05 Minnetonka Rhododendron flower along Tranquility Court in the Franklin Farm section of Oak Hill, Fairfax County, Virginia Minnetonka Rhododendron flower along Tranquility Court in the Franklin Farm section of Oak Hill, Fairfax County, Virginia.	
	IT→T	What is this plant named after?		Kalmia. Kalmia is a genus of about ten species of evergreen shrubs from 0.2-5 m tall, in the family Ericaceae. They are native to North America ... saw it during his travels in Carolina, and after his return to England in.	-
	IT→I	Is shiny and silver with shorter sleeves and fit and flare.		-	
	IT→IT	Is this plant poisonous?		Heracleum mantegazzianum, commonly known as giant hogweed, is a monocarpic perennial herbaceous plant in the carrot family Apiaceae ...These serious reactions are due to the furanocoumarin derivatives in the leaves, roots, stems, flowers, and seeds of the plant. Consequently, it is considered to be a noxious weed in many jurisdictions.	

Table 14: Data examples in diffierent task type. Due to the limitations of the table, we have cropped the displayed text.

Domain	Candidate Image	Candidate Text	FLUX Image	Google Image	Query Text
animal		The golden poison frog is the most poisonous animal on the planet; these frogs produce deadly alkaloid batrachotoxins in their skin glands as a defense against predators. To become poisoned a predator generally must attempt to consume the frog, ... has modified sodium channels unaffected by batrachotoxin.			What is the primary defense mechanism of this animal?
architecture		Neoclassical buildings are characterized by their magnificence of scale, the prominent use of columns, the use of geometric forms and symmetry. ...Samriddhi Bhavan,...National library of India, Kolkata			What are some examples of this style in Indian public buildings?
artwork		"Finding Peace Under Pressure: A Close Look at the new Butterfly of Peace". The Houston Museum of Natural Science. Retrieved 2021-07-05. "Aurora Butterfly of Peace on Display at Smithsonian". The Gemmological Association of Great Britain. Retrieved 2021-07-05.			Where was this display shown?
currency		The euro was founded on 1 January 1999, when it became the currency of over 300 million people in Europe. For the first three years of its existence it was an... Slovenia joined the Eurozone in 2007, Cyprus and Malta in 2008, Slovakia in 2009, Estonia in 2011 and Latvia on 1 January 2014.			When did this currency become available?
entertainment		Thomas Middleditch as Richard Hendricks, a coder and founder/CEO of Pied Piper.T.J. Miller as Erlich Bachman (seasons 1-4), an Chris Diamantopoulos as Russ Hanneman ...a brash, loud and fiery billionaire investor who provides Pied Piper with their Series A.			Who is the CEO of this company in the TV series Silicon Valley?
food		An Italian beef sandwich features thin slices of seasoned roast beef, dripping with meat juices, on a dense, long Italian-style roll, believed to have originated in Chicago, where its history ... Despite the name, it is almost completely unknown in Italy.			What city is this sandwich believed to have originated in?
language		In the early 6th century BCE, the Neo-Babylonian Empire conquered the ancient Kingdom of Judah, destroying much of Jerusalem and exiling its population far to the East in Babylon. During ...details on Hebrew and Aramaic in the gospels.)			What languages were spoken in this region during the Roman period?
literature		The Adventures of Huckleberry Finn (1973), by Robert James Dixon – a simplified version Big River: The Adventures of Huckleberry Finn, a 1985 ... Classics imprint was released in November 2017.			What form of media was this book adapted into in 1985?
mythology		Throughout India, on contemporary poster art, Ganesha is portrayed with Sarasvati (goddess of culture and art) or Lakshmi (goddess of luck and prosperity) or both. Ganesha, Lakshmi and Sarswati ... to be the brother of Sarasvati and Lakshmi.			What is the relationship between this deity and Sarasvati in Maharashtra?
organization		During World War II, ARC operated the American Red Cross Clubmobile Service to provide servicemen with food, entertainment and "a connection home." In a ...During the Vietnam War 627 American women served in the ARC Supplemental Recreation Overseas Program. At the invitation			What service did this organization provide to boost soldier morale during the Vietnam War?
person		Runnels later re-emerged in 1998, under her real name, as the on-screen girlfriend of Val Venis. When Runnels claimed to be pregnant with Venis' baby, he dumped her... broke up by July, when Jacqueline Moore became frustrated with Runnels' infatuation with Meat.			Who did this person claim to be pregnant with in 1998?
pharmaceutical		DHA-paclitaxel (or Taxoprexin) is an investigational drug (from Protarga Inc) made by linking paclitaxel to docosahexaenoic acid (DHA), a fatty acid that is easily ...may be able to treat more types of cancer than Taxol has been able to treat.			What is the advantage of this drug over paclitaxel?
plant		The species was first described as Salpiglossis integrifolia by William Jackson Hooker in 1831. It was transferred to the genus Petunia as P. integrifolia by Hans Schinz and Albert Thellung... ranges, with P. inflata growing in more northern areas.			What was the original genus of this plant?
sport		The Columbia University Lions are the collective athletic teams and their members from Columbia University, an Ivy League institution in New York City, United States. The current director of athletics is Peter Pilling.			What is the name of the athletic teams from this university?
vehicle		A specialized Lexus LS 460 is used in a warehouse-sized driving simulator at Toyota's Higashifuji Technical Center in Shizuoka, Japan. This vehicle is mounted ... automotive safety features in a secure environment.			What is the purpose of this driving simulator at Toyota's Higashifuji Technical Center?

Table 15: Examples of synthetic data. Due to space constraints, we have abbreviated the displayed content.


```
>> SYSTEM
You are a helpful assistant.

>> USER
Provide an Entity along with a Passage introducing this entity.
Generate a concise Description of the entity's appearance. This
description, limited to fewer than 25 words, will be used to create an
image of the entity.
Example1:
Entity:
Heracleum mantegazzianum
Passage:
<passage>
Description:
Heracleum mantegazzianum: a tall plant with large, compound leaves and
white, umbrella-like flower clusters.

Entity:
<Entity>
Passage:
<passage>
Description:
```

Figure 9: Fused-Modal Data Synthesis Step 3 Prompt.

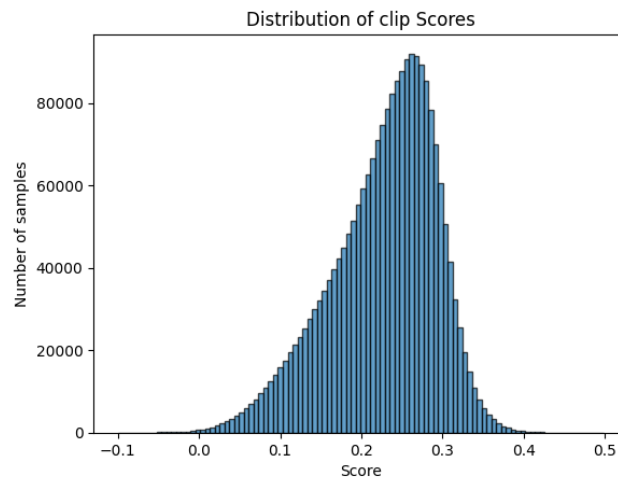


Figure 10: The distribution of relevance scores for images retrieved by Google, along with their corresponding captions.

testing processes of our GME. Consequently, performance in other languages could not be evaluated.